

DATA STRUCTURES - 2025

SECTION – A

1. Define abstract data type.
2. Infer the usage of multilists.
3. Recall when an overflow and underflow condition occur during stack operations.
4. List some applications of queues.
5. Indicate the properties of binary tree.
6. Distinguish binary search tree and AVL tree.
7. Define B+ Tree with an example.
8. Express how a graph differs from a spanning tree with an example.
9. Distinguish internal and external sorting.
10. State hashing function.

SECTION – B

11. A. Infer how lists and linked list are stored in memory with an example and write a function to check current position is last in a linked list.
B. Distinguish circular linked list and doubly linked list during insertion and deletion operation.
12. A. Express an algorithm to add two polynomials having n terms respectively using linked list with an example.
B. Explain how linked list are used for sorting the following set of numbers using radix sort. A = [432, 8, 530, 90, 88, 231, 11, 45, 677, 199]
13. A. Outline the pseudo-code for performing the following operations in stacks using array implementation.
 - (1) Create
 - (2) Push
 - (3) Pop
 - (4) Stack TopB. Identify how stacks are used for checking whether an expression is balanced or not for every right and left brace, bracket and parenthesis.
14. A. State the procedure for performing enqueue, dequeue, Isempty, and Isfull operations in a queue data structure with example data.
B. Recognize the differences between circular queue and dequeue.
15. A. Classify the three types of binary tree traversal and write the pseudo-code for in-order traversal with an example.
B. Explain how a postfix expression is converted into an expression tree with an example.

SECTION – C

16. A. Demonstrate how minimum cost spanning tree can be obtained by Prim's method with an algorithm.
B. Trace the above algorithm for the following graph and analyze the time and space complexity.

